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Department of Electronics and Communication Engineering

19EAC384 – Computer Networks Lab (S6 EAC)

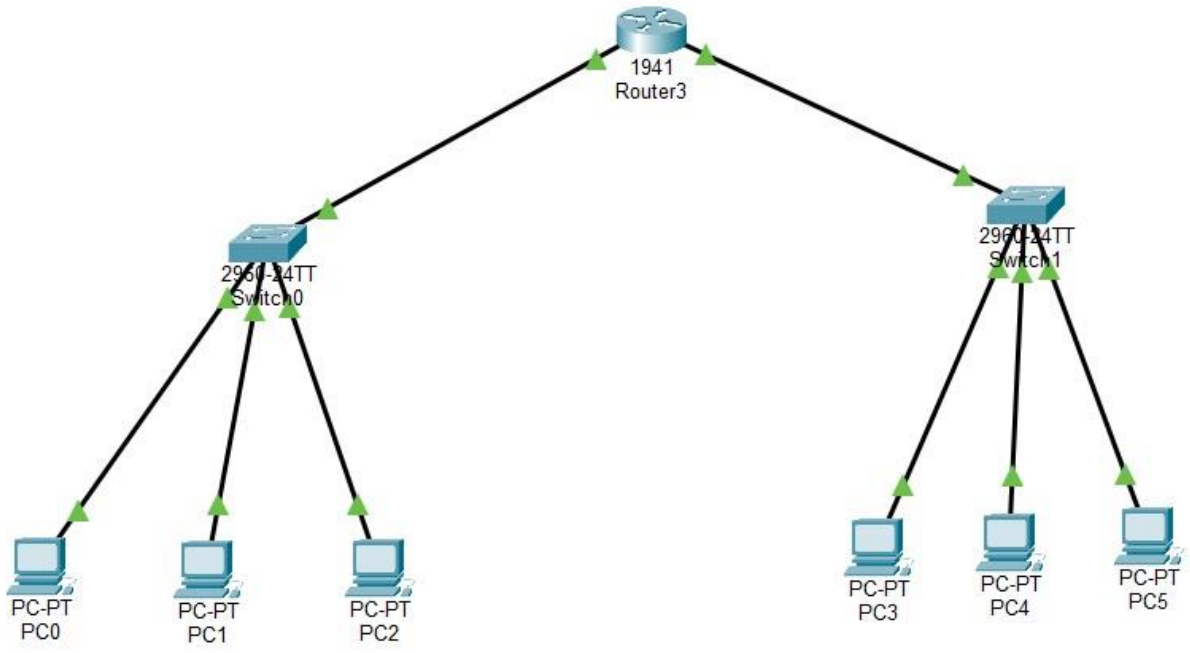
LABSHEET 4 - Assignment

BUILDING A WIDE AREA NETWORK (WAN) SET-UP

Aim:

- To build a WAN set up by configuring computers to form Local Area Networks (LAN) and establish communication between PCs in different LANs using Routers.
- To assign IP address dynamically to PCs using DHCP server.

Exercises-1: Build two simple LANs using 3 machines connected via star topology. Establish communication between two PCs belonging to different LANs using a single Router. In your document, highlight the steps adopted to set up the individual LANs and to configure the router and PCs to establish successful communication.



CLI Commands Listed:

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#! Assign IP address to interface
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.173.11.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#! (Optional) Set another IP for second interface if needed
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 192.173.22.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#end
Router#
Router#! Save the configuration
Router#copy running-config startup-config
Destination filename [startup-config]?
%SYS-5-CONFIG_I: Configured from console by console
```

Ping Commands:

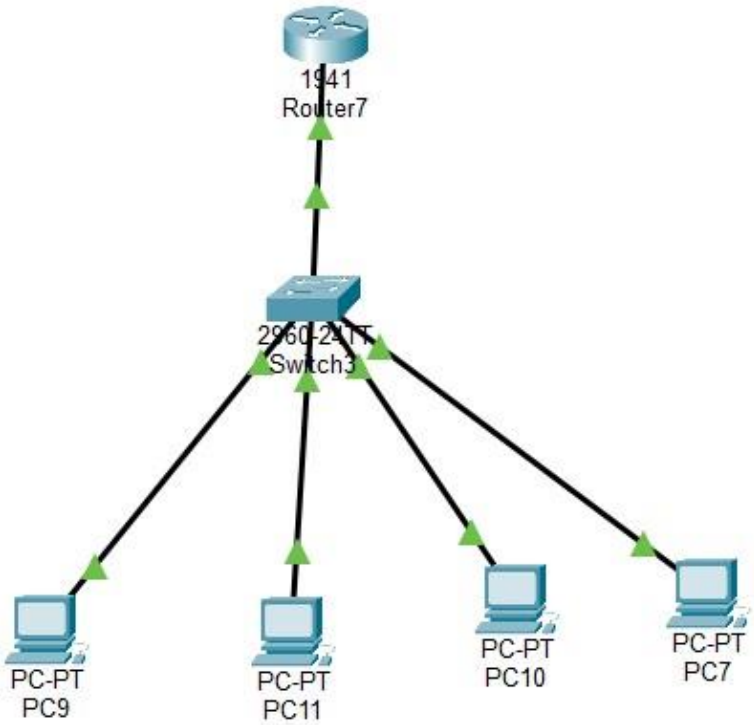
```
C:\>ping 192.173.22.2

Pinging 192.173.22.2 with 32 bytes of data:

Reply from 192.173.22.2: bytes=32 time<1ms TTL=128
Reply from 192.173.22.2: bytes=32 time<1ms TTL=128
Reply from 192.173.22.2: bytes=32 time=6ms TTL=128
Reply from 192.173.22.2: bytes=32 time=6ms TTL=128

Ping statistics for 192.173.22.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 3ms
```

```
C:\>ping 192.173.11.4

Pinging 192.173.11.4 with 32 bytes of data:

Reply from 192.173.11.4: bytes=32 time<1ms TTL=128
Reply from 192.173.11.4: bytes=32 time=4ms TTL=128
Reply from 192.173.11.4: bytes=32 time=4ms TTL=128
Reply from 192.173.11.4: bytes=32 time=3ms TTL=128

Ping statistics for 192.173.11.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 2ms
```

Inference:

This lab activity reinforced the importance of correct IP addressing and routing configuration in enabling successful communication between different networks. It highlighted that even minor misconfigurations can disrupt connectivity, and emphasized the need for systematic troubleshooting and verification of network parameters such as IP addresses, subnet masks, and default gateways. Overall, the hands-on experience deepened my understanding of how routers facilitate communication across LANs and enhanced my confidence in configuring and managing network setups.

Exercises-2: To configure a DHCP server in Cisco Packet Tracer that assigns IP addresses dynamically to 3 PCs connected to a switch. In your document give your observations.



CLI Commands Listed:

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.173.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#ip dhcp excluded-address 192.173.1.1 192.173.1.9
Router(config)#ip dhcp pool LAN_Pool
Router(dhcp-config)#network 192.173.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.173.1.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Ping Commands:

```
C:\>ping 192.173.1.11

Pinging 192.173.1.11 with 32 bytes of data:

Reply from 192.173.1.11: bytes=32 time<1ms TTL=128
Reply from 192.173.1.11: bytes=32 time<1ms TTL=128
Reply from 192.173.1.11: bytes=32 time<1ms TTL=128
Reply from 192.173.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.173.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 192.173.1.10

Pinging 192.173.1.10 with 32 bytes of data:

Reply from 192.173.1.10: bytes=32 time=11ms TTL=128
Reply from 192.173.1.10: bytes=32 time=5ms TTL=128
Reply from 192.173.1.10: bytes=32 time=3ms TTL=128
Reply from 192.173.1.10: bytes=32 time=4ms TTL=128

Ping statistics for 192.173.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 11ms, Average = 5ms
```

Inference:

This activity provided practical insight into the process of dynamic IP address allocation using DHCP. By configuring the DHCP settings via the command line, I gained a clearer understanding of how routers manage IP distribution within a network. The initial challenges in selecting the correct interface and setting up the DHCP pool enhanced my problem-solving skills and deepened my comprehension of network automation. Overall, the exercise boosted my confidence in using the CLI and improved my understanding of DHCP functionality in Cisco Packet Tracer.